

Performance-Focused Design Exercise Packet

Part of ASME IDETC 2025 Workshop:

Adventures in Tradespace Exploration

17 August 2025

Group ID: 3050



Annotate, mark-up, and write on this page as needed.

Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables.

134 in	83.1 in	80 in	18.1 in	706 lbf	5.95:1	49 in	1010 Kwh 25	0.512 in	1.75 in
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Do you think you could have found a better solution given more time?

Yes, we took all the time available.

What, in this round, stopped you from finding a better solution?

Time + lack of Humvee comparison.

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Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables.

181 in	81.2 in	69.2 in	9.7 in	655 lbf	2.37:1	32 in	1070 kWh	.877 in	1.99 in
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Do you think you could have found a better solution given more time?

Yes, with more time we could gather more information for a better solution.

What, in this round, stopped you from finding a better solution?

More information about different aspects would've allowed us to find a better solution.

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Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables

199 in	84.2 in	68.2 in	22.7 in	419 lbf	2.4:1	39 in	5020 kWh	0.331 in	1.74 in
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Do you think you could have found a better solution given more time?

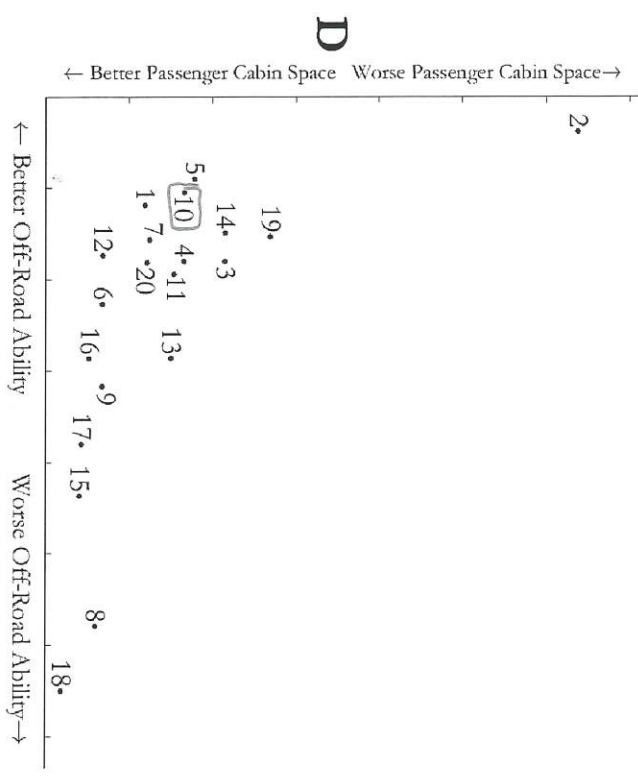
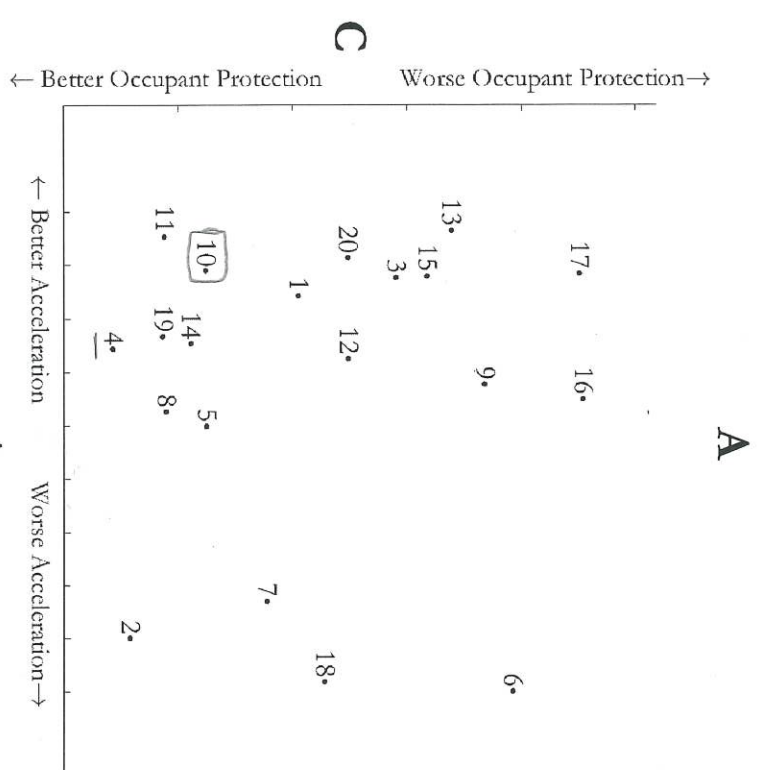
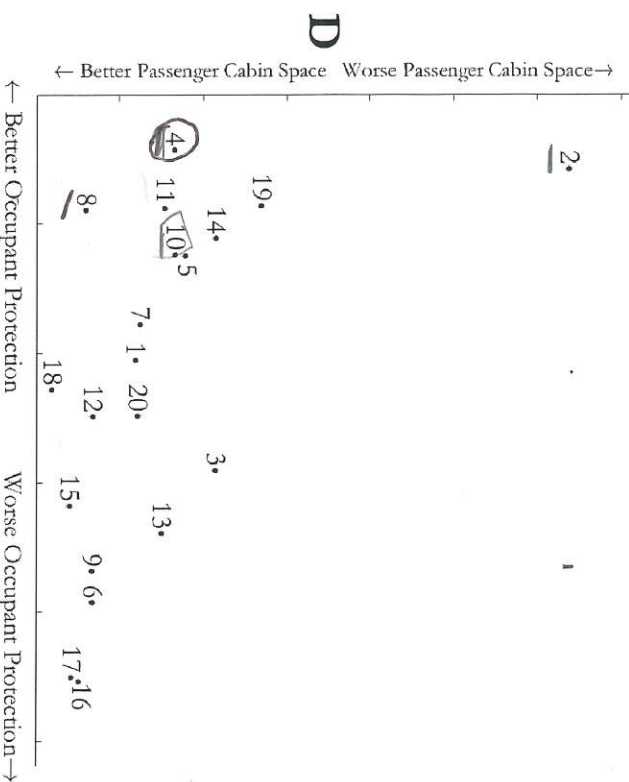
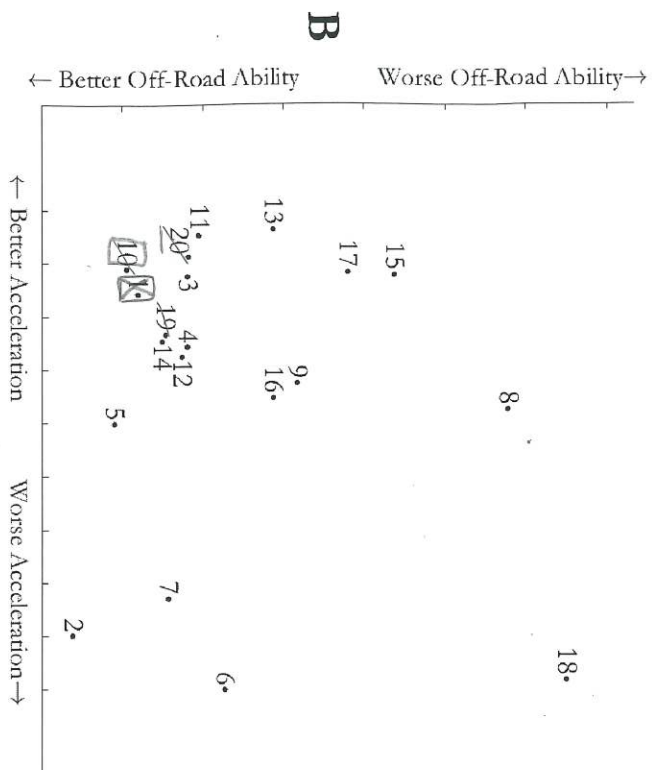
Yes, given more time we can find a better solution.

What, in this round, stopped you from finding a better solution?

Time prevented us from finding a better solution.

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A

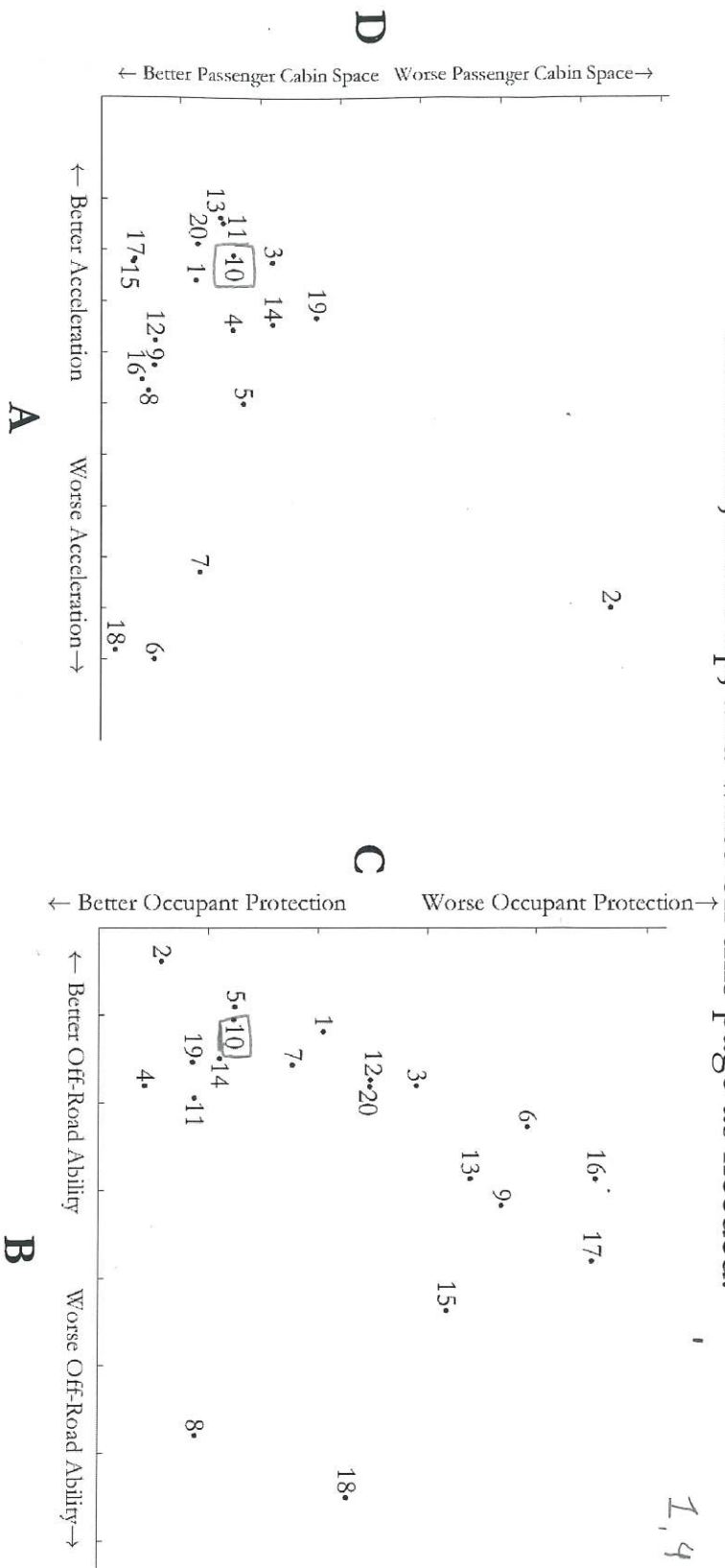
A

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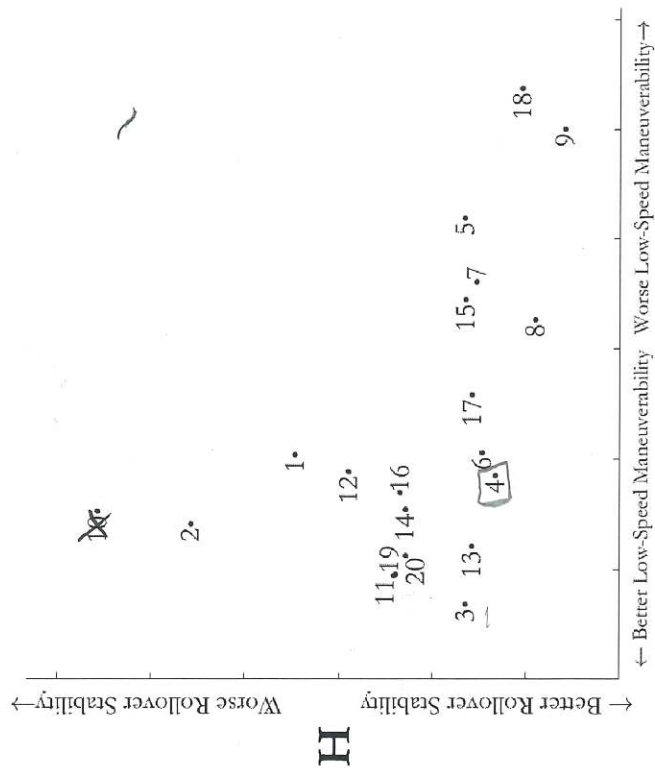
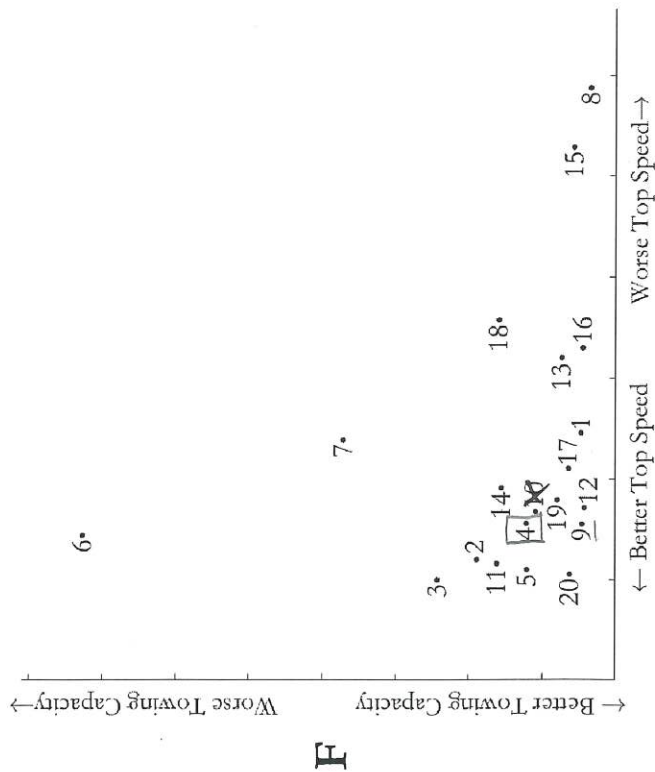
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E

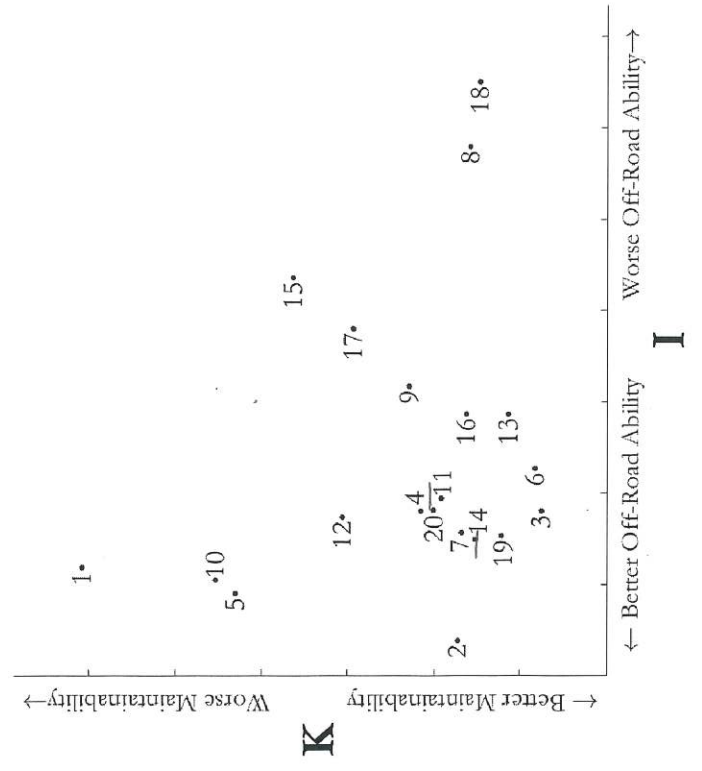
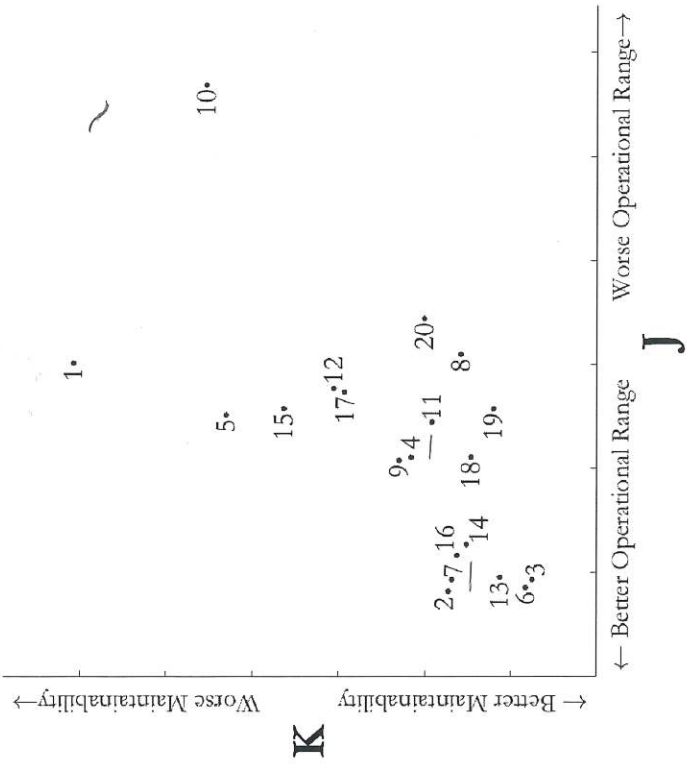
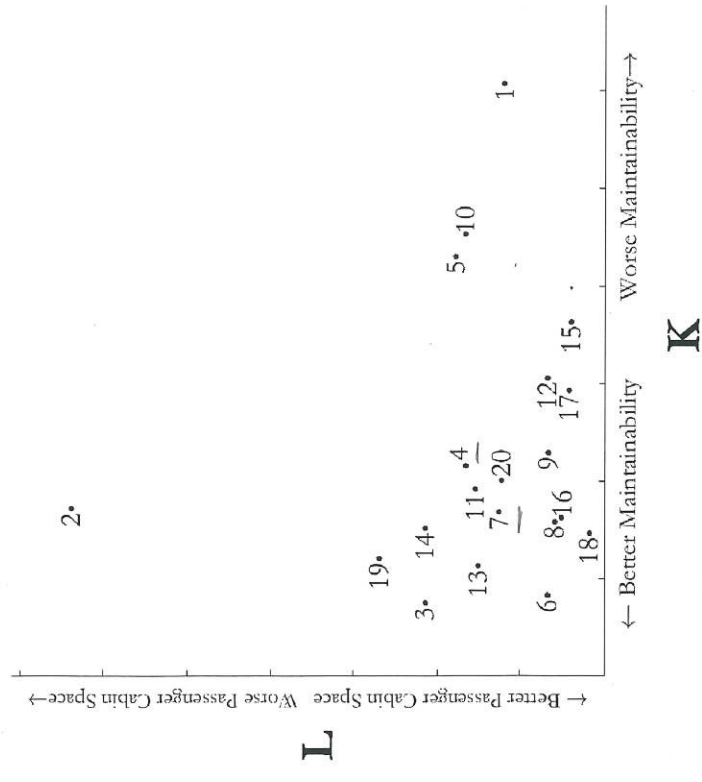
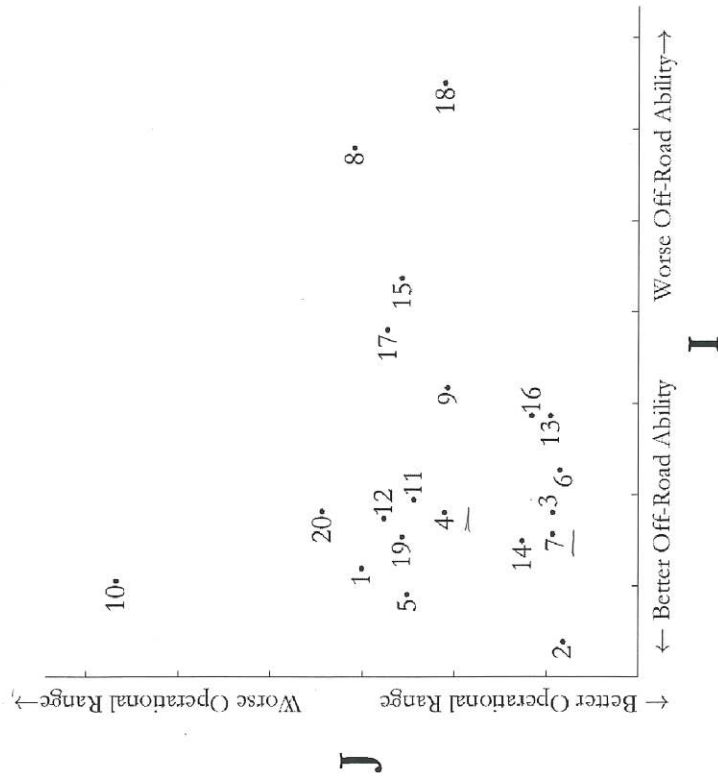
G

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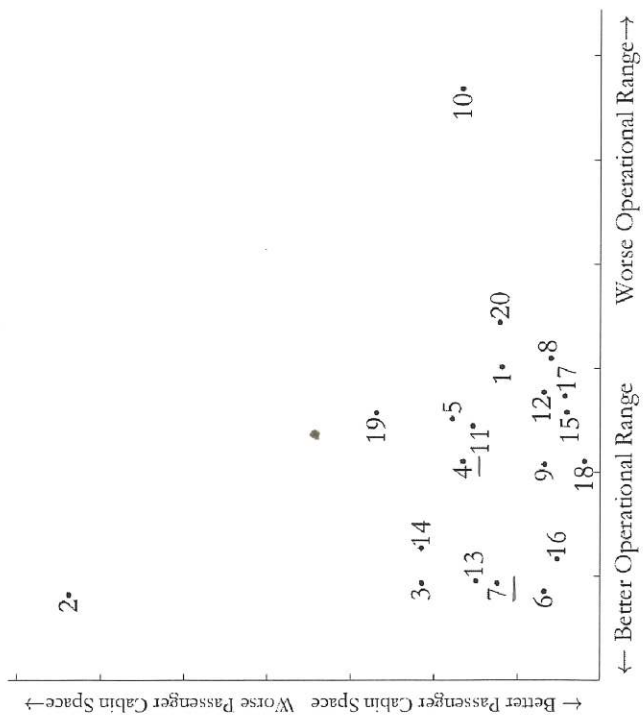
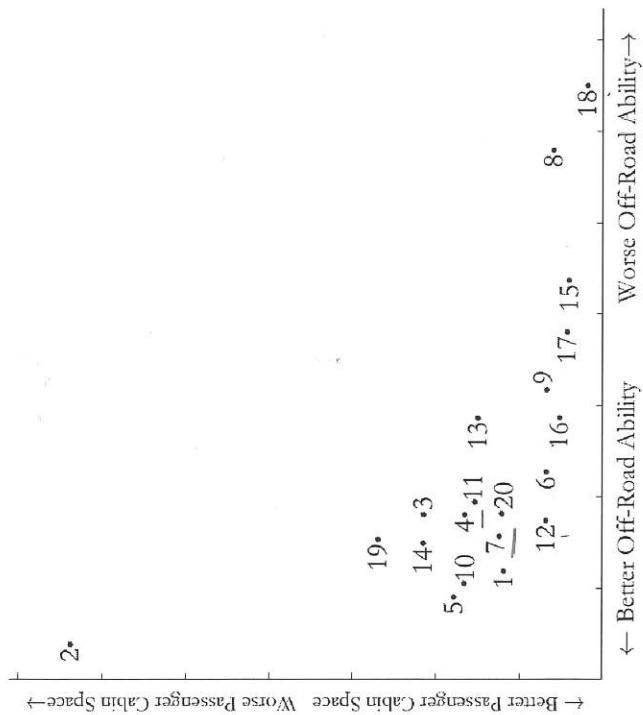
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Annotate, mark-up, and write on this page as needed.



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